The formula, term by term

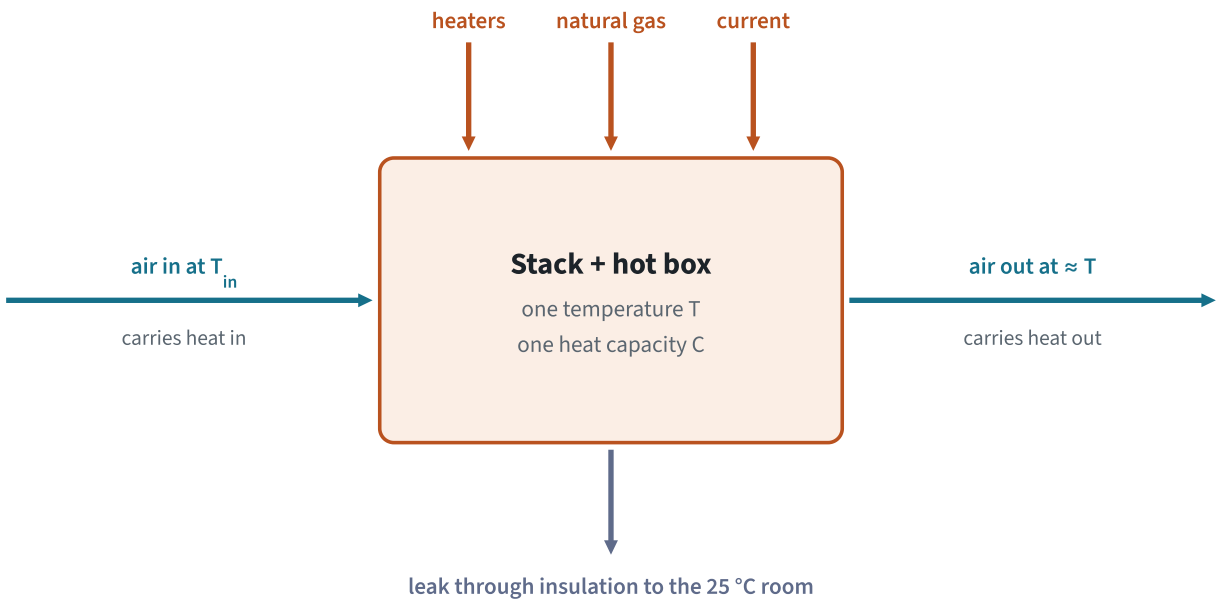
The twin decides how fast the stack warms or cools with one rule from school physics: **heat cannot appear or vanish**. Every simulated minute it adds the heat coming in, takes away the heat going out, and the leftover changes the stack temperature.

The stack and its insulated hot box are treated as one "tank of heat" with a single temperature. Five things fill or drain the tank. Each has one number, and all of those numbers were measured from the plant's own data. They are not taken from a textbook.

THE FORMULA IN THE TWIN (WARMING SPEED, °C PER HOUR)

$$dT/dt = a \cdot k \cdot (T_{in} - T) + 7.62 + 0.370 \cdot NG + 6.61 \cdot I - 0.0121 \cdot (T - 25$$

T = stack temperature (°C). T_{in} = temperature of the air entering the stack. NG = natural-gas flow in normal litres per minute (ln/min: litres measured at 0 °C and normal air pressure). I = current per string in amps. a = (air flow ÷ 6000)$^{1.5}$. k = 0.561 when the air warms the stack, 0.316 when it cools it.



One tank, five flows. The air term can go either way. The three heat sources only add heat. The loss only removes it.

What each term means

TERM	NUMBER IN THE TWIN	PH ME
Air explained →	0.561 /h (warming) or 0.316 /h (cooling) × (air ÷ 6000)$^{1.5}$ × (T_{in} − T)	Air hea wh hot tha sta tak ou it is coo Mo mo mo hea
Heaters explained →	+7.62 °C/h while the heaters are on	Ele hea the bo fix am hea
Natural gas explained →	+0.370 °C/h for each ln/min of gas	Pa gas ch en rel as nea sta
Current explained →	+6.61 °C/h for each amp per string	Ce ma ele are pe the sh

TERM	NUMBER IN THE TWIN	PH ME
		co as
Loss explained →	−0.0121 /h × (T − 25 °C)	He thr the ins Th the the it l

How to read the pages

- Page 2: why one tank of heat is a fair picture, how heat in kilowatts turns into °C per hour, and how big the tank is.
- Page 3: the air term, why it has two speeds, and the 1.5 power on the air flow.
- Page 4: where the air-in temperature comes from (heat recovery and the 40 kW air heater).
- Page 5: the heaters, the gas, the current and the leak, checked against chemistry and electricity.
- Page 6: where the stack settles in each stage, with a calculator you can play with.

IF YOU REMEMBER ONE THING

Stack warming speed = air term + heaters + gas + current − leak. Each term has one number fitted on plant data.

Source: 05_twin_2d/app.js v34 (constants refitted 2026-11-05 by 03_data_analysis/scripts/twin_fits.py). Physics checks on these pages are hand calculations made for this explainer. Written 2026-11-05.

Why a tank of heat works

The formula is the law of energy conservation written for one lump of material. The lump is the stack plus its hot box. It has one temperature and one size of "heat tank".

From the law to the formula

Energy conservation says: the heat stored in the lump changes by exactly the heat that flows in minus the heat that flows out. The stored heat is heat capacity × temperature, so:

ENERGY BALANCE OF ONE LUMP

$$C \cdot dT/dt = P_{air} + P_{heaters} + P_{gas} + P_{current} - P_{leak}$$

C = heat capacity of the lump: the energy needed to warm it by 1 °C (kJ per °C). Each P is a heat flow in kilowatts. dT/dt = how fast the temperature changes.

Divide both sides by C and every heat flow becomes a warming speed in °C per hour. That is why every number in the twin is written in °C per hour. The fit never needed to know C. It only measured how fast the plant's stack warmed or cooled.

THE SAME BALANCE DIVIDED BY C

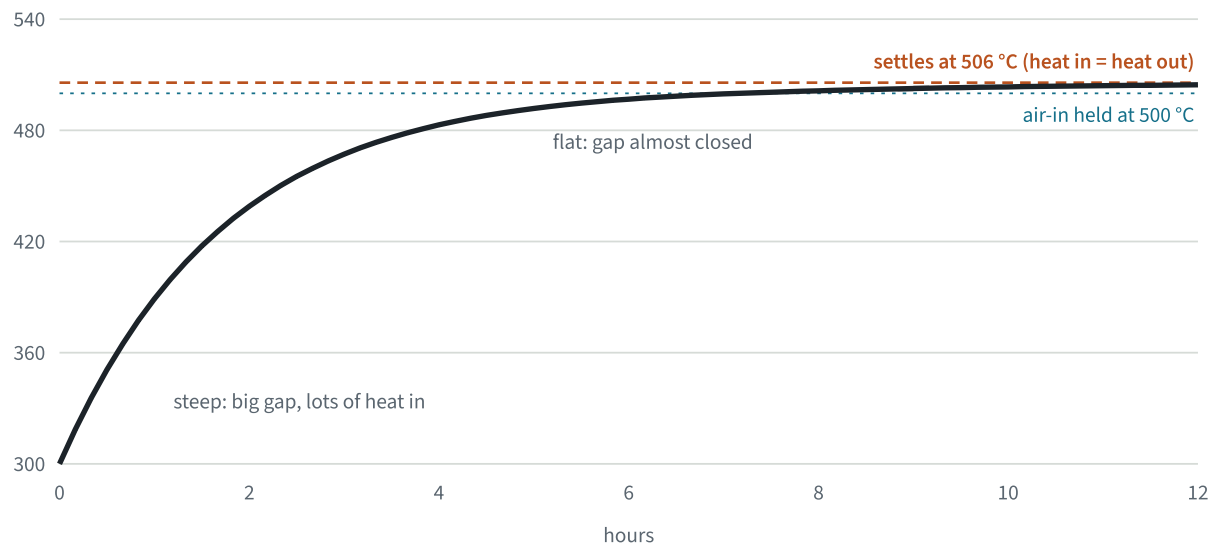
$$dT/dt = P_{air}/C + P_{heaters}/C + P_{gas}/C + P_{current}/C - P_{leak}/C$$

For example, "heaters = +7.62 °C/h" means $P_{heaters} \div C = 7.62$ °C per hour.

Why one lump is a fair picture

Inside the hot box, heat spreads between stack, metal frames and gas far faster than the plant's settings change. Stage changes and air-heater ramps take tens of minutes to hours. So most of the time the inside of the box moves together, and one temperature describes it well enough. The twin uses the stack's own thermocouples as that temperature.

The picture is not perfect. The insulation and the heavy heat exchangers warm and cool more slowly than the stack. That is the reason the air term needs two numbers (page 3).



The model run with hot-standby settings from 300 °C. A big gap between air and stack fills the tank fast; as the gap closes the warming slows. It takes about 1.8 h to cover two thirds of the way. This "fast then slow" curve is the fingerprint of a heat balance.

How big is the tank?

The data gives only speeds, not C itself. The air term lets us estimate it, because we know how much heat the air can carry:

STEP	CALCULATION	RESULT
Air mass flow at 6000 ln/min	`6000 ÷ 60 000 m³/s × 1.293 kg/m³`	`0.129 kg/s`
Heat the air carries per °C	`0.129 kg/s × 1.050 kJ/(kg·°C)`	`0.136 kW per °C`
Tank size when warming	`0.136 × 3600 s/h ÷ 0.561 /h`	`≈ 0.87 MJ per °C`
Tank size when cooling	`0.136 × 3600 ÷ 0.316`	`≈ 1.55 MJ per °C`
Same as a block of steel of	`871 kJ/°C ÷ 0.5 kJ/(kg·°C)`	`≈ 1.7 tonnes`

> **ASSUMPTION**
> This assumes the air leaves the stack at the stack temperature. If it leaves a bit cooler, the real tank is smaller. So read 0.87 MJ per °C as an upper estimate. On page 5 it is used to turn °C per hour into kilowatts.

One step per simulated minute

The twin works out the warming speed, multiplies it by one minute (1/60 h) and adds the result to the stack temperature: **T_{new} = T + speed × 1/60**. This simple step is accurate because the fastest the tank ever reacts is roughly one hour, which is 60 times longer than the step.

WORKED MINUTE

T = 480 °C, T_{in} = 500 °C, air 6000, heaters on

speed = 0.561 × 20 + 7.62 − 0.0121 × 455 = 13.33 °C/h, so the stack gains 0.222 °C in that minute.

IF YOU REMEMBER ONE THING

Heat in minus heat out, divided by the size of the tank, gives the warming speed. The twin only needs the speeds, and the plant data supplies them.

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The air term

Air is the biggest lever on the stack temperature. Air hotter than the stack heats it; air cooler than the stack cools it. The twin writes this as **$a \cdot k \cdot (T_{in} - T)$**: a measure of the air flow, times a speed number, times the temperature gap.

The physics: air is a heat carrier

A stream of air carries heat equal to **mass flow × heat capacity × temperature**. Air enters at T_{in} and, after passing the hot cells, leaves at roughly the stack temperature T. The heat it hands over is therefore:

HEAT DELIVERED BY THE AIR

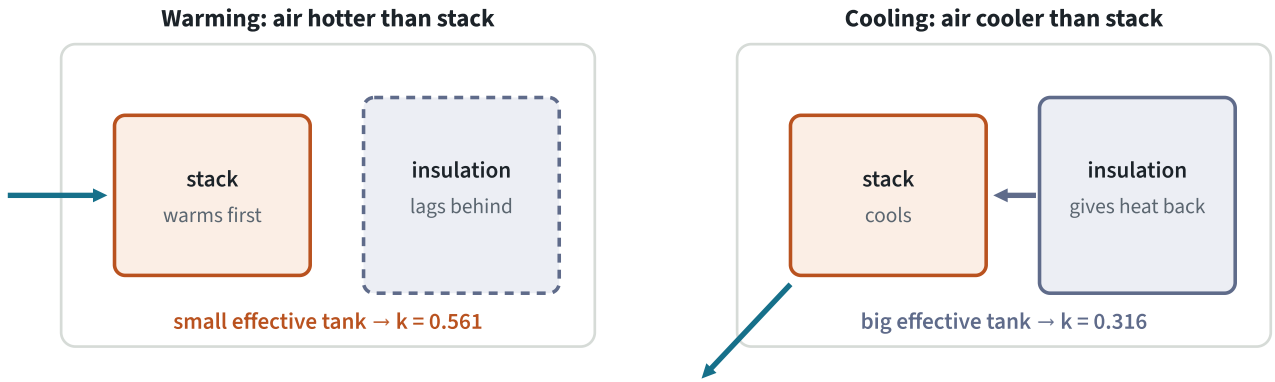
$$P_{air} = \dot{m} \cdot c_p \cdot (T_{in} - T)$$

$\dot{m}$ = air mass flow (kg/s); c_p = heat capacity of air, about 1.05 kJ per kg per °C at these temperatures. Divided by the tank size C this becomes $k \cdot (T_{in} - T)$, so $k = \dot{m} \cdot c_p / C$.

At 6000 ln/min the air carries 0.136 kW per °C. A 100 °C gap is therefore about 13.6 kW, much more than the heaters (about 1.8 kW, page 5). The fitted k = 0.561 per hour means: at 6000 ln/min the air closes 56 % of the gap per hour, if nothing else acted.

Why two speeds: 0.561 warming, 0.316 cooling

A single tank would give the same k both ways. The plant does not behave like that. It cools more slowly than a single tank predicts. The reason is the insulation and the heavy metal around the stack. They form a second, slower tank.

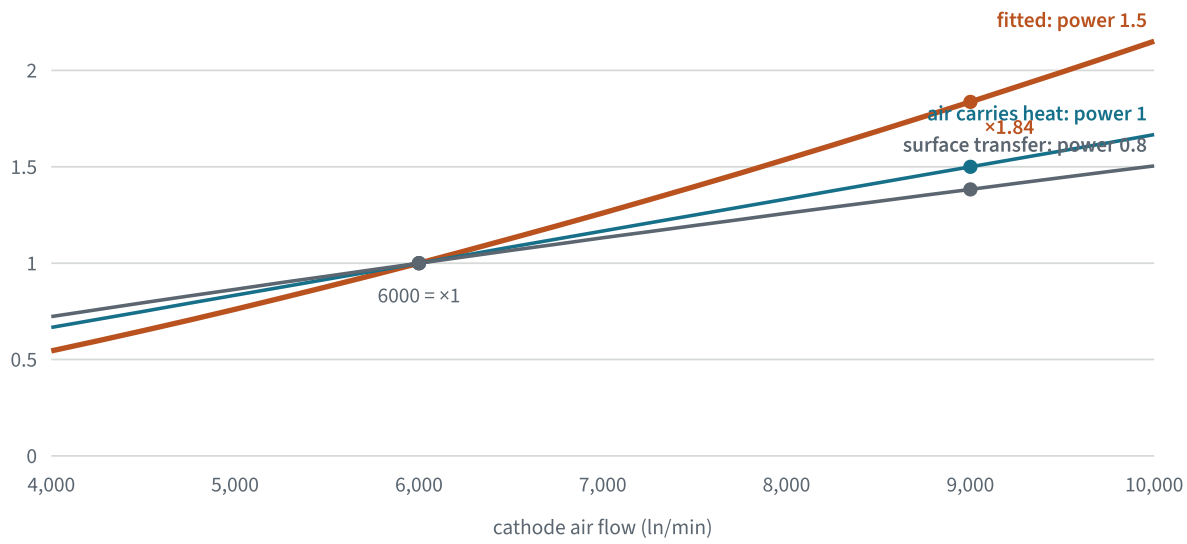


When the stack is warmed, the insulation has not caught up yet, so the air only has to heat the stack: a smaller tank, faster change. When it is cooled, the warm insulation feeds stored heat back, so the air must remove that too: a bigger tank, slower change.

Instead of modelling two tanks, the twin uses one tank with two speeds. In tank sizes that means about 0.87 MJ per °C when warming and 1.55 MJ per °C when cooling. This is a short cut that matches the plant, not a law of nature.

Why the air flow has a power of 1.5

The term $a = (\text{air} \div 6000)^{1.5}$ says how much more heat extra air moves. Simple physics expects a power of 1: double the air, double the heat carried. If the air does not fully reach the stack temperature, the power is even a bit lower, around 0.8. That is how heat transfer from a surface usually grows with flow.



At 9000 ln/min the fitted rule moves 1.84 times the heat of 6000 ln/min. Simple physics would say 1.50 (power 1) or 1.38 (power 0.8).

WHAT 1.5 MEANS

The plant cools faster at high air flow than simple physics explains. The single number is most likely also absorbing effects the one-tank picture leaves out. More air also cools the insulation and the heavy heat exchangers. It lowers the heat recovery (page 4). And the plant used high air flow mainly when the stack was hot. Treat 1.5 as measured, not explained.

A second, fitted cooling curve after hot standby

When the plant leaves hot standby with the heaters off, the twin switches to a cooling curve fitted on five real cool-downs:

COOLING SPEED AFTER HOT STANDBY (°C/H)

$$dT/dt = -f \cdot 0.0605 \cdot (air \div 6000)^{1.65} \cdot u \cdot (u \div 475)^{0.5}, \quad u = T -$$

u is the stack's height above room temperature. f ramps from 0 to 1 over about 8 minutes, because the plant takes a few minutes to reach full cooling speed.

The physics shape behind it: a plain leak (Newton's law of cooling) would make the speed proportional to u. The fitted curve grows with $u^{1.5}$, so it cools **faster than Newton when hot**. Two real effects push in that direction. Heat radiation between hot parts rises very steeply with temperature, and hot gas carries and conducts heat better. Below 300 °C the twin blends back into the normal heat balance, because the plant runs never went below about 260 °C.

STACK	AIR LN/MIN	FITTED COOLING CURVE °C/H	HEAT BALANCE °C/H
500 °C	6,000	-28.7	-23.8
500 °C	9,000	-56.1	-52.6
300 °C	6,000	-12.7	-13.8
300 °C	9,000	-24.7	-30.5

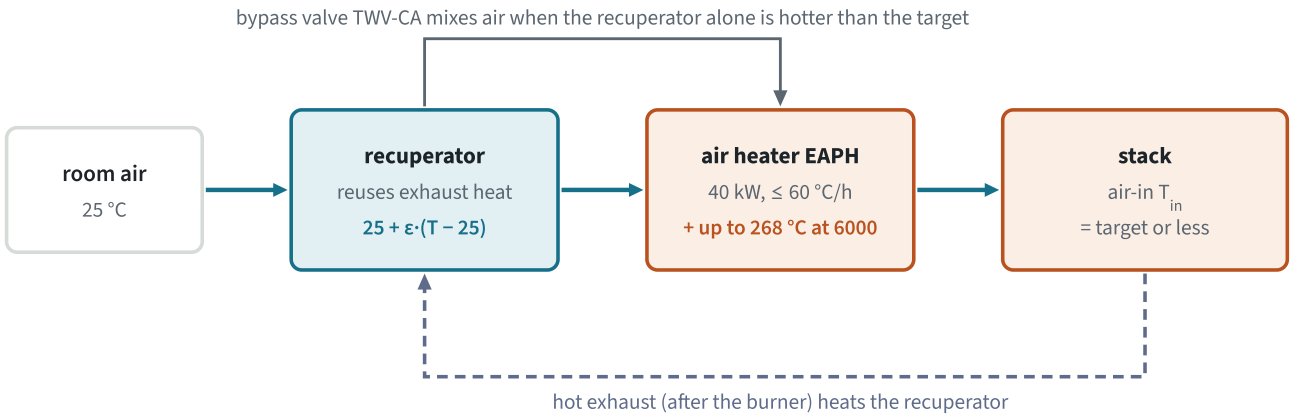
IF YOU REMEMBER ONE THING

Air moves heat in proportion to the temperature gap. Two speeds stand in for the slow insulation; the 1.5 power is measured, not explained.

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Where the air-in temperature comes from

The air term needs the air-in temperature T_{in}. The twin calculates it from two pieces of equipment: the **recuperator**, which reuses heat from the exhaust, and the **electric air heater**, which adds up to 40 kW on top.



Room air is warmed twice before it reaches the stack. The recuperator recovers exhaust heat; the air heater makes up the rest, up to its 40 kW and its 60 °C per hour ramp.

1. Heat recovery: the recuperator

A recuperator is a heat exchanger. Hot exhaust passes on one side of a metal wall and cold fresh air on the other. Engineers rate one by its **effectiveness ε**: how much of the largest possible warming the air actually gets.

AIR LEAVING THE RECUPERATOR

$$T_{rec} = 25 + \varepsilon \cdot (T - 25), \quad \varepsilon = 0.88 \text{ at } 6000 \text{ ln/min} \rightarrow 0.83 \text{ at } 8000 \text{ ln/}$$

The exhaust is taken as roughly as hot as the stack. Example: stack 600 °C at 6000 ln/min gives 531 °C air.

Why ε falls with more air: the exchanger's wall area stays the same while more air rushes through, so each kilogram of air spends less time touching the hot wall. Textbooks call this a lower "number of transfer units". The fall from 0.88 to 0.83 is

that effect. These two values were set by hand, because they beat the fitted values on the plant replays.

2. The electric air heater (EAPH)

While the heaters are on, the air heater lifts the recuperator air up to the plant's target. Typical targets are 300 °C at cold start, 500 °C in hot standby, 580 °C in fuel switch and 600 °C in part load. Three limits from the real plant apply:

- **Power.** 40 kW, of which about 91 % reaches the air. The most it can lift the air is power ÷ ($\dot{m} \cdot c_p$).
- **Speed.** The plant's control system ramps the target at 60 °C per hour, so the twin's air-in rises no faster.
- **No cooling.** A heater cannot cool. If the recuperator air is already hotter than the target, the bypass valve mixes in fresh air and holds the target, as on the plant.

> **PHYSICS CHECK THAT WORKS OUT**
> Largest lift at 6000 ln/min = 40 kW × 0.909 ÷ 0.136 kW per °C = **268 °C**. The twin uses 268 °C (RISE_6000), and divides it by the air flow ratio at other flows: at 9000 ln/min the lift is 179 °C. Number and physics agree.

When the heaters are off or the air heater has nothing to add, T_{in} follows the recuperator with a 10-minute delay. The twin moves it a fraction $1 - e^{-1/10}$ of the remaining gap each minute. This is the standard "first-order lag" for a part with its own small heat store.

3. The air heater's power reading is a true energy balance

ELECTRIC POWER SHOWN FOR THE AIR HEATER

$$P = (\text{air} \div 60\,000) \cdot 1.293 \cdot c_p \cdot (T_{out} - T_{rec}) \div 0.909, \quad c_p = 1003 +$$

air ÷ 60 000 turns ln/min into m^3/s; 1.293 kg/m^3 is air density at 0 °C; c_p rises slightly with temperature; 0.909 = share of electric power that ends up in the air, fitted on the plant (89–91 %). Capped at 40 kW.

HOT STANDBY EXAMPLE	VALUE
Stack (settled)	506 °C
Recuperator air 25 + 0.88 × (506 − 25)	448 °C
Air-in target	500 °C
Heat put into the air	7.1 kW
Electric power shown (÷ 0.909)	7.8 kW

> **SMALL KNOWN GAP**
>
> The c_p formula gives about 1.05 kJ/(kg·°C) at 450 °C. Air tables give about 1.07. The 0.909 efficiency was fitted with the twin's own c_p, so it absorbs most of this 2–3 % difference.

A second recuperator model, for display only

The air-heater inlet reading uses a richer recuperator model. It lags the stack by about 32 minutes, because the exchangers are heavy. It also adds 0.85 °C per ln/min of gas, because with gas on the exhaust burner runs hotter. It was tested inside the heat balance too. There it made the 9000-air cool-down replay miss by 156 °C instead of 74 °C, so the stack balance keeps the simple version.

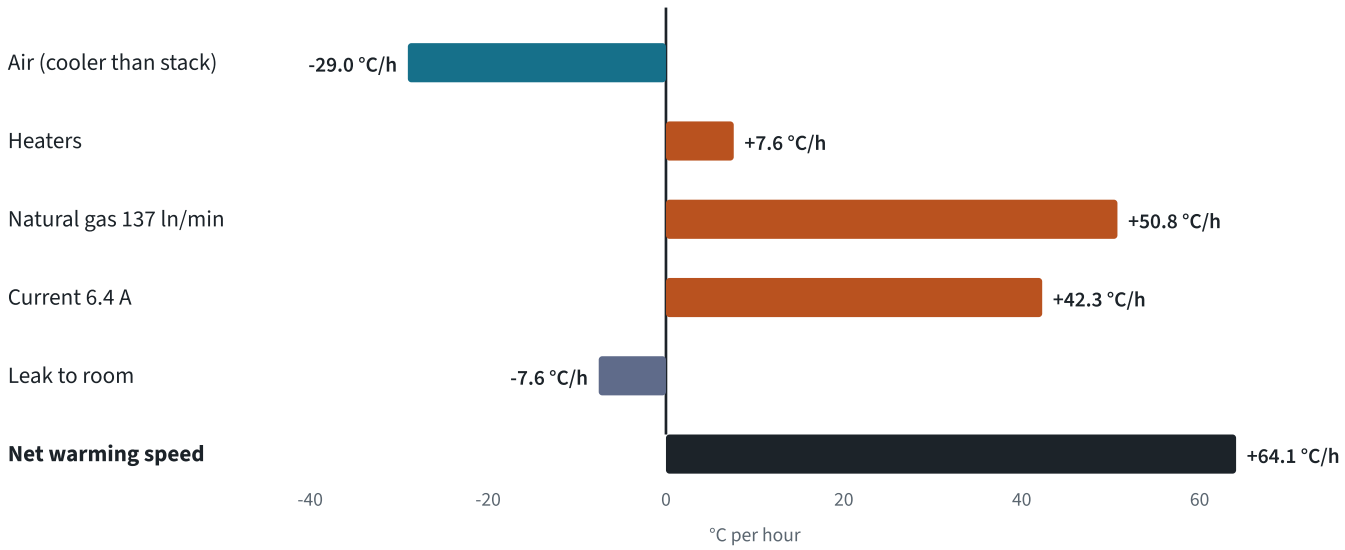
IF YOU REMEMBER ONE THING

Air-in = recovered exhaust heat + what the 40 kW heater can add, no faster than 60 °C per hour. The 268 °C lift matches 40 kW of air heating.

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Heat added and heat lost

Three terms only add heat: the hot-box heaters, the natural gas and the electric current. One term only removes heat: the leak through the insulation. Each is a straight line in the twin: a fixed number times the gas flow, the current or the temperature.



Part load, stack at 650 °C, air-in 600 °C, air 9000 ln/min, gas 137 ln/min, 6.4 A (plant medians). Gas and current dominate; the air already cools the stack. The net +64 °C per hour is why the twin needs a "stack keeper" in long part load (page 6).

Turning °C per hour into kilowatts

Multiplying by the tank size from page 2 (0.87 MJ per °C, an upper estimate) gives each term in kilowatts. This lets us check the numbers against chemistry and electricity.

TERM	TWIN NUMBER	AS HEAT	AT A TYPICAL POINT
Heaters	7.62 °C/h	≈ 1.8 kW	whenever the heaters are on
Natural gas	0.370 °C/h per ln/min	≈ 90 W per ln/min	80 ln/min: +29.6 °C/h ≈ 7.2 kW
Current	6.61 °C/h per A	≈ 1.60 kW per A	6.4 A: +42.3 °C/h ≈ 10.2 kW
Leak	0.0121 /h × (T − 25)	≈ 2.9 W per °C	500 °C: −5.7 °C/h ≈ 1.39 kW

Heaters: a fixed push

The hot-box heaters (the off-gas and reformer heaters around the stack) add **+7.62 °C per hour**, about 1.8 kW net into the tank. The 40 kW air heater is **not** in this term. Its heat arrives through the air-in temperature (page 4). The twin treats the hot-box heaters as simply on or off at a typical power, which matches how they ran in the data.

Natural gas: chemistry

One ln/min of natural gas (taken as methane) carries about **0.60 kW** of chemical energy: 35.8 MJ per m^3 ÷ 60 000. The fitted 0.370 °C/h per ln/min equals about 90 W, so **about 15 %** of the gas energy ends up warming the tank. Where does it come from with no current flowing?

- **The reformer.** The gas is mixed with air (the POx air) and partly burned: $CH_4 + ½\ O_2 \rightarrow CO + 2\ H_2$ releases heat. More oxygen burns more of it.
- **The exhaust burner.** Unused fuel burns after the stack. Most of that heat leaves with the exhaust, but some reaches the hot box through walls and the recuperator.

STAGE (PLANT MEDIANS)	OXYGEN PER METHANE	SHARE OF GAS ENERGY RELEASED IN THE REFORMER
Fuel switch: 80 ln/min gas, 343 ln/min POx air	0.90	≈ 29 %
Part load: 137 ln/min gas, 312 ln/min POx air	0.48	≈ 4 %

Air is 21 % oxygen. Pure partial oxidation needs 0.5 oxygen per methane; full burning needs 2. In fuel switch the plant feeds about 0.9, so roughly 29 % of the gas energy is released already in the reformer. The fitted 15 % reaching the tank fits inside that.

WATCH THE SPLIT IN PART LOAD

The gas number was fitted in fuel switch, where no current flows. In part load the plant runs the reformer leaner (0.48 oxygen per methane), so physics says the reformer releases much less heat there. The current number was then fitted "on top of the gas". So in part load only the **sum** of gas + current is backed by data; how it splits between them is not.

Current: electrochemistry

A fuel cell turns the fuel's energy into electricity and heat. The total energy available per unit of charge is the **thermoneutral voltage**, about 1.25 V per cell for hydrogen. The cell delivers its working voltage, roughly 0.8–0.9 V. The difference becomes heat:

HEAT MADE BY THE CELLS

$$P_{heat} = I \cdot (1.25\ V - V_{cell}) \times \text{number of cells in series}$$

Heat grows in step with the current, which is why the twin uses a straight line: 6.61 °C/h per amp per string, about 1.6 kW per amp.

OPEN CHECK

Closing this check needs the exact cell count and the working cell voltage in part load. The data gives neither directly (the min-cell tag suggests 119 cells per stack). Two things also make the twin's straight line approximate. Resistance heat grows faster than the current, and internal reforming of fuel absorbs heat. The plant ran only at about 6.4 A, so the line is reliable near that current only.

Leak: heat through the insulation

Heat leaks through the insulation in proportion to the temperature difference with the room (Newton's law of cooling): **−0.0121 per hour × (T − 25)**. In plain terms the insulation passes about 2.9 W per °C, so about 1.4 kW at 500 °C. With everything off, the tank would lose 63 % of its extra heat in 1 ÷ 0.0121 = 83 hours.

CHECK AGAINST THE PLANT

With the air off at 200 °C the formula gives −0.0121 × 175 = **-2.1 °C per hour**. The plant was measured at about −2 °C per hour. The leak number was fitted on exactly those air-off stretches, the cleanest data there is: only the leak acts.

IF YOU REMEMBER ONE THING

Gas and current are the big heat sources in power stages, the heaters a small fixed push, and the leak a slow drain of about 1.4 kW at 500 °C.

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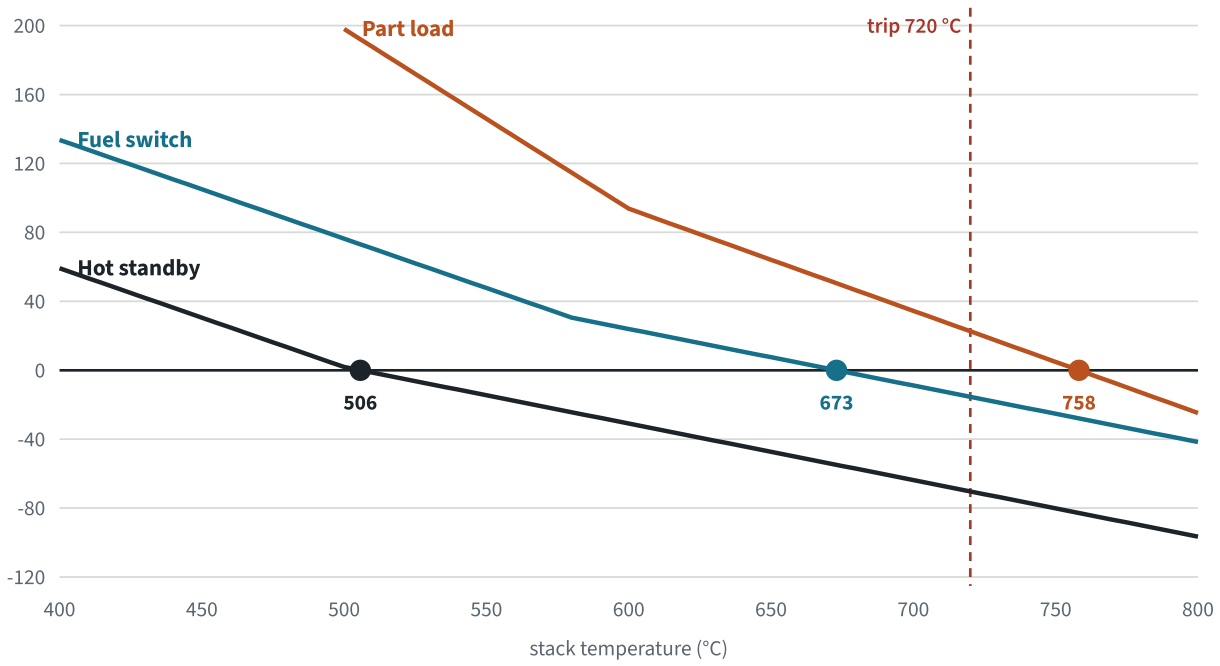
Where the stack settles

The stack stops warming where heat in equals heat out. Set the formula to zero and you get the temperature each stage drives the stack towards. For hot standby it lands right where the plant sits.

SETTLE POINT (WARMING SPEED = 0, AIR-IN HELD AT ITS TARGET)

$$T_{settle} = (a \cdot k \cdot T_{in} + heaters + gas + current + 0.0121 \cdot 25) \div (a \cdot k + 0.$$

A weighted average: the air pulls the stack towards T_{in}, the leak pulls towards 25 °C, and the heat sources push it above both.



Each line is the warming speed at plant-median settings, with air-in held at the stage's target. Above zero the stack warms, below zero it cools, so every stage pushes the stack towards its dot. The bend in each line is where the air switches from warming (0.561) to cooling (0.316).

STAGE	SETTINGS (PLANT MEDIANS)	SETTLE POINT	PLANT
Hot standby	air 6000, air-in 500 °C, heaters on	506 °C	sits at about 505 °C ✓
Fuel switch	+ gas 80 ln/min, air-in 580 °C	673 °C	lasts about 10 min, so it never gets there: it warms about +22 °C/h (twin starts near +30)
Part load	air 9000, gas 137, 6.4 A, air-in 600 °C	758 °C	never ran longer than 44 min; 758 °C would be past the 720 °C trip

That last row is the reason for the twin's **stack keeper**. Above 660 °C it raises the air, then lowers the air-in target. This is control logic added to the twin, not physics. The plant data never shows how the plant itself holds a long part load.

Try it: one minute of the heat balance

What is not physics in the twin's stack curve

- **Smooth stage changes.** When a stage changes, the knobs jump in one step, but the plant turns over minutes. The twin blends the old warming speed into the new one along a smooth S-curve, $10u^3 - 15u^4 + 6u^5$ (a "minimum-jerk" curve), timed from the plant's own stage borders. This stands in for parts the one tank leaves out: heater coils, gas lines and valves that have their own small heat stores.
- **Stage rules and the stack keeper.** These are operating rules, not physics.
- **The other 96 sensors.** They are looked up from plant history by stage and stack temperature, not calculated.

Where the physics stops

LEFT OUT	EFFECT
Second (insulation) tank	Replaced by two air speeds; slow after-effects of long heating or cooling are only roughly right
Reformer and burner chemistry	Gas heat is one straight line; in part load only gas + current together is backed by data
Cell voltage model	Current heat is a straight line, reliable near 6.4 A only
Temperature spread inside the stack	One temperature stands for the whole stack; hot and cold spots are not shown
The 1.5 power on air	Measured, not explained; outside 6000–10 000 ln/min it is a guess

IF YOU REMEMBER ONE THING

Where heat in equals heat out, the stack stops. Hot standby settles at 506 °C, as on the plant; long part load would pass the trip, so the twin adds a keeper.

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